



Ratnesh Kumar Singh

Electronics & Communication Engineer

Networking • AI Automation • UI/UX

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SKILLS & TOOLS

NETWORKING & IT
TCP/IP, DNS, Routing & Switching, Cybersecurity
PROGRAMMING
C, C++, Python, HTML, CSS, API Integration
AI & AUTOMATION
n8n, Google AI Studio, OpenAI, AI Agents
EMBEDDED & ENG.
Arduino, MATLAB, Tinkercad, Verilog, ArcGIS
UI/UX & DESIGN
Figma, Wireframing, Prototyping, Canva

LANGUAGES

English (Professional)
Hindi (Native)
Bengali (Conversational)

INTERESTS

Photography, Sketching, Audio & Video Editing, Tech Trends

PROFILE SUMMARY

Motivated Electronics & Communication Engineering graduate (2026) with a versatile skillset spanning networking, IT infrastructure, AI automation, and UI/UX design. Experienced in developing embedded hardware prototypes, designing user interfaces in Figma, and programming in Python/C++. Strong understanding of networking concepts (TCP/IP, routing), V2X communication, and workflow automation. Eager to contribute to technology-driven roles across networking, software development, or systems integration. Open to relocation across India.

KEY PROJECTS

Adaptive Smart Antenna System (V2X) (MATLAB • V2X • ECE Hardware)

Designed and developed an adaptive smart antenna system prototype to improve wireless communication reliability in high-mobility vehicular environments. Implemented beamforming algorithms that dynamically direct signals toward moving vehicles, helping reduce interference and maintain stable connectivity. Supporting emerging V2X (Vehicle-to-Everything) standards.

Simple AI Assistant (Python • APIs • Conversational AI)

Created an intelligent assistant using Python and API integrations. Implemented features like natural language command understanding, conversational response generation, and real-time information retrieval, demonstrating client-server communication and data handling skills.

Smart Parking System (Arduino • Ultrasonic Sensors • IoT)

Designed an IoT prototype using Arduino and ultrasonic sensors to detect available parking spaces in real-time, demonstrating practical sensor integration and network communication.

UI/UX Design & Portfolio Website (Figma • HTML • CSS • Vercel)

Designed multi-screen UI layouts and interactive prototypes in Figma, enforcing accessibility and typography guidelines. Developed and deployed a clean, responsive portfolio website on Vercel.

EDUCATION

Qual.	Institution	Board/University	Year	Specialization
B.Tech	Swami Vivekananda Inst. of Sci. & Tech.	MAKAUT (W.B.U.T.)	2022 - 2026	ECE
12th Std	P.B.S College	B.S.E.B	2020	Science (PCM)
10th Std	Vidyanjali High School (H.S.)	I.G.C.S.E	2018	Science

CERTIFICATIONS & TRAINING

Python Programming 1-Month Course	C Language Programming 1-Month Course
Electric Vehicle Service Tech. 1-Month Training	GIS (Geographic Info. Systems) 2-Weeks Training

EXTRA-CURRICULAR ACTIVITIES

- HAM Radio:** Workshop on Innovation & General Awareness (2025).
- BSNL Telecom:** Completed a one-day industrial visit at BSNL Telecom (2025).
- Cyber Security:** Workshop 'Empower Yourself With Cyber Awareness' (2025).